

1. What is the purpose of a fragment shader in the graphics pipeline?
 - a) To perform transformations on vertices
 - b) To apply textures to polygons
 - **c) To perform pixel-level calculations**
 - d) To perform lighting calculations
2. Which of the following is not a common technique for optimizing rendering performance?
 - a) Level of detail (LOD) rendering
 - b) Occlusion culling
 - **c) Post-processing effects**
 - d) Vertex buffer objects (VBOs)
3. Which Python library provides a higher-level interface for working with OpenGL than PyOpenGL?
 - **a) Pyglet**
 - b) Pygame
 - c) Panda3D
 - d) PyOpenCL
4. What is the purpose of a Render Target in graphics frameworks?
 - a) To store the current window size
 - **b) To create a texture for rendering to**
 - c) To manage the rendering order of objects
 - d) To apply post-processing effects
5. Which of the following is a common technique for creating realistic lighting in 3D graphics?
 - **a) Phong shading**
 - b) Gouraud shading
 - c) Flat shading
 - d) Wireframe rendering
6. Which component of a material in a graphics framework defines the appearance of an object's surface?
 - a) Vertex shader
 - b) Fragment shader
 - **c) Texture**
 - d) Uniform variable
7. Which of the following is not a common post-processing effect?
 - a) Motion blur
 - b) Depth of field
 - c) Ambient occlusion
 - **d) Specular reflection**

1. Question: Which OpenGL primitive type is used to draw a solid 3D cube?

- A) GL TRIANGLES
- **B) GL QUADS**
- C) GL LINES
- D) GL POINTS

2. Question: What is the purpose of a vertex array object (TAO) in OpenGL?

- A) To store texture coordinates
- B) To store vertex positions
- C) To manage shader programs
- **D) To optimize rendering performance**

3. Question: In OpenGL, what does the term "frustum" refer to?

- A) A type of shader
- **B) The visible portion of the scene within the camera's view**
- C) A texture mapping technique
- D) A lighting model

4. Question: Which transformation matrix is used to apply perspective projection in OpenGL?

- A) Model matrix
- B) View matrix
- **C) Projection matrix**
- D) Normal matrix

5. Question: What is the purpose of a framebuffer object (FBO) in OpenGL?

- A) To store depth information
- B) To manage shaders
- **C) To render offscreen**
- D) To handle user input events

1. Which of the following lighting effects can enhance the three-dimensional nature of a scene?
 - a) Ambient lighting
 - b) Specular highlights
 - c) Bump mapping textures
 - d) All of the above
2. Shading in a rendered image allows viewers to observe the 3D nature of a shape without the need for:
 - a) Vertex colors
 - **b) Textures**
 - c) Lighting effects
 - d) Shadows
3. The Lambert model and the Phong model are commonly used illumination models. The Phong model is more computationally intensive than the Lambert model because it includes:
 - a) Ambient lighting
 - b) Diffuse lighting
 - **c) Specular lighting**
 - d) Emissive lighting
4. The angle at which a ray of light meets a surface is calculated as the angle between the direction vector of the light ray and a:
 - **a) Normal vector**
 - b) Tangent vector
 - c) Reflection vector
 - d) Refraction vector
5. Which shading model uses vertex normal vectors and calculates the effect of light at each vertex in the vertex shader?
 - a) Flat shading model
 - b) Gouraud shading model
 - **c) Phong shading model**
 - d) Lambert shading model
6. The base Light class in the graphics framework stores data that could be needed by any type of light and includes a constant that specifies the type of light. What are the three types of lights represented by this constant?
 - a) Ambient, diffuse, and specular
 - **b) Directional, point, and spot**
 - c) Ambient, directional, and point
 - d) Emissive, reflective, and refractive
7. Which type of light object simulates rays of light being emitted from a single point in all directions and incorporates attenuation?
 - a) Ambient light
 - b) Directional light
 - **c) Point light**
 - d) Spotlight
8. In the graphics framework, which class represents the ambient light and only uses the color data?
 - a) Light
 - b) Object3D
 - **c) AmbientLight**
 - d) DirectionalLight

What is OpenGL?

• OpenGL is a widely adopted cross-platform application programming interface (API) that enables developers to write programs for graphics processing units (GPUs). It allows for efficient rendering of 2D and 3D graphics, making it a popular choice for creating interactive visual applications!

1. Texture -1:

In computer graphics, a texture is a 2D image that is applied to a 3D surface. This image provides visual detail and surface properties like color, pattern, and material appearance.

2. Why Light Models?

Light models are crucial for simulating how light interacts with objects, adding realism to computer graphics. They create shadows, reflections, and surface scattering effects, making objects appear more natural and conveying their shape, form, and spatial relationships.

3. Light Model Factors:

Light models are built on factors that determine how light interacts with surfaces. Key factors include:

- Light Source Properties: Color, intensity, and direction.
- Surface Properties: Diffuse reflection (scattering), specular reflection (mirror-like), and ambient reflection (indirect light).
- Attenuation: Light intensity decreases with distance (distance attenuation).

Examples of Light Models:

- Phong Shading: Combines diffuse and specular reflection.
- Blinn-Phong Shading: An optimized version of Phong shading.
- Lambert Shading: Focuses on diffuse reflection.
- Ray Tracing: A complex model simulating light ray paths.